

Employee Attrition Prediction – Executive Summary

Project Objective

The objective of this project was to develop a machine learning solution to predict employee attrition and identify the key factors influencing employees to leave the organization. By analyzing historical HR data, the project aims to support data-driven decisions that improve employee retention and workforce planning.

Approach

The project involved data cleaning, preprocessing, exploratory data analysis, feature encoding, and training three machine learning models: Logistic Regression, Random Forest Classifier, and Gradient Boosting Classifier. The models were evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC Score to identify the most effective predictive model.

Key Findings

- The employee attrition rate is approximately **16%**, indicating that a minority of employees leave the organization each year.
- Important factors influencing attrition include **Monthly Income, Overtime, Age, Total Working Years, Job Role, and Years at Company**.
- The best-performing machine learning model demonstrated strong capability in identifying employees at risk of leaving, making it a valuable decision-support tool for HR teams.

Recommendations

- Monitor employees identified as having a high risk of attrition.
- Improve work-life balance by reviewing overtime practices.
- Strengthen employee engagement and career development initiatives.
- Review compensation strategies for employees with lower income levels.
- Incorporate predictive analytics into HR planning to proactively address employee retention.

Conclusion

This project demonstrates how machine learning can help organizations identify attrition risks before employees resign. By leveraging predictive analytics, HR departments can implement targeted retention strategies, reduce turnover costs, and improve overall workforce stability.